Modified Locally Linear Embedding based on Neighborhood Radius

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Abstract-**As a nonlinear dimensionality reduction technology, locally linear embedding is a kind of very competitive approach with good representational capacity for a broader range of manifolds and high computational efficiency. However, LLE and its variants determine the neighborhood for all points with the same neighborhood size, without considering the unevenly distribution or sparsity of data manifold. This paper presents a new performance index-ratio of neighborhood radius to predict the unevenly distribution or sparsity of data manifold, and a new approach that dynamically determines the neighborhood numbers based on the ratio of neighborhood radius, instead of adopting a fixed number of nearest neighbors per data point. This approach has clear geometry intuition as well as the better performance, compared with LLE algorithm. The conducted experiments on benchmark data sets validate the proposed approach.**

I. INTRODUCTION

Most real data lies on a low dimensional manifold embedded in a high dimensional space. The task of manifold learning is to recover the meaningful low-dimensional structures hidden in high dimensional data. Classical techniques, such as Principal Component Analysis (PCA)[1] and Multidimensional Scaling (MDS)[2], are efficient to find the true structure of the data when dealing with a linear manifold. These methods are theoretically simple and easy to implement. However, an assumption has been taken in these methods: the data lies in a linear or almost linear sub space of the high-dimensional space and the embedding can be obtained using these linear methods. Obviously, this assumption is too restrictive as many real data cannot satisfy the linear assumption.

Recently, a new class of nonlinear embedding techniques has been designed to discover the structure of high-dimensional data that lies in a nonlinear high-dimensional manifold and find their embedding in a low dimensional Euclidean space. Some of the more prominent examples of nonlinear manifold learning algorithms are ISOMAP[3], Locally Linear Embedding (LLE)[4] and Laplacian Eigenmaps [5]. Among these, LLE is an effective nonlinear algorithm. Compared with others, the notable advantages of LLE are: i) only two parameters should be set;ii) it can reach global minimization without local minima; iii) it can well preserve the local geometry of high dimensional data in the embedded space.

The primary idea of the LLE algorithm is to preserve the relationships between neighbors in manifold data and map the high dimensional data in low dimensional Euclidean space. Assuming that each data point lies on a locally linear patch of a manifold, LLE characterizes local geometry by representing each data as an approximate affine mixture of its neighbor points. The local geometry of each patch can be described by the reconstruction weighs with which a data point is reconstructed from its nearest k neighbors.

The LLE algorithm was demonstrated on a number of artificial and real world data sets. It has many variants, such as Hessian LLE[6], incremental LLE[7], supervised LLE[8,9], etc. However, LLE and its variants are based on the assumption that the whole data manifold is evenly distributed. These approaches determine the neighborhood for all points with the same neighborhood size, without considering the uneven distribution or sparsity of data manifold. To solve this problem, this paper proposes a new performance index-ratio of neighborhood radius to predict the uneven distribution or sparsity of data manifold. Based on the ratio, this paper also proposes an improved LLE algorithm that can dynamically selects the neighborhood numbers for each point on manifold according to the structure of the manifold.

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The remainder of the paper is organized as follows. Section 2 proposes an improved LLE algorithm based on the definition of ratio of neighborhood radius. In section 3 experimental results on synthetic data sets are presented. Finally, section 4 gives the conclusion.

II. Dynamical Selection of Neighborhood Numbers

LLE largely depends on the neighborhood structure. It always assumes that the whole data manifold is evenly distributed, and determines the neighborhood for all points with the same neighborhood size. However, the assumption that the real world data is evenly distributed is not always true, and always unevenly distributed or sparse, as shown in Figure 1. This makes the selection of same neighborhood size for all points unsuitable. In Figure 1, the distribution of data in high dimension space is unevenly and sparse. The data can be divided into two groups, and there are some outliers between the two groups data. For this data, when applying the original LLE algorithm, the property of uneven distribution of data is not considered and it will select same neighborhood numbers for all points. Such as, LLE selects 6 neighbor points for the point x_1, x_2 and x_3, respectively. In fact, x_3 is an outlier, and should be deleted in the procedure of reconstruction. For this shortcoming of LLE, we propose a new method to dynamically select the neighborhood numbers, based on the definition of neighborhood radius and the ratio of neighborhood radius.

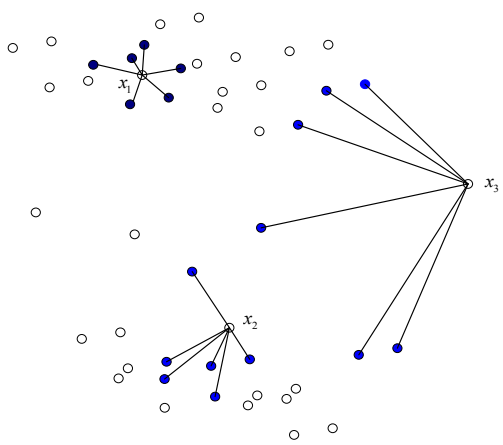

Figure 1: Neighbors selection of LLE algorithm. The neighborhood numbers of all the points is 6.

Definition 1: the neighborhood radius of point X_i is defined as follows:

$$radius(X_i) = \frac{1}{K}\sum_{j=1}^{k} disp(X_i, X_{ij})$$

Where K is the number of neighbors, X_{ij} is the K nearest neighbors, and $disp(X_i, X_{ij})$ is the Euclidean distance between X_i and X_{ij}.

Definition 2: the ratio of neighborhood radius is defined as follows:

$$ratio(X_i) = \frac{\left|radius(X_i) - radius(X_j)\right|}{radius(X_i)} \qquad i \neq j$$

The two definitions can be used as a performance index to predict the non-uniformity or sparsity of dataset, and identify the outliers. Under the ideal condition, the ratio of neighborhood radius for even distribution dataset should be equal to zero. With the ratio value increasing, the non-uniformity of dataset became serious. If the ratio value is greater than a threshold, we can think the point is an outlier. Based on this idea, these two definitions can be used to dynamically select the neighborhood numbers according to the data distribution. The method is described as follows:

Step 1. Find the data set of k nearest neighbors for each point X_i by Euclidean distances;

Step 2. Compute the neighborhood radius and the ratio of neighborhood radius for each point X_i, if the ratio is greater than a constant *C*, it is that some point in the neighborhood of X_i is too far away from X_i, and it may be an outlier point, so, delete it from neighborhood of X_i.

Step 3. Repeat step 2. If the neighborhood numbers is less than a constant α, then this point is an outlier itself, delete this point from the original data.

Using this method, the neighborhood numbers is not fixed, and can be dynamically selected according to the ratio of neighborhood radius. For the three points in Figure 1, set their neighborhood radius as r_1, r_2 and r_3, respectively. After